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Explaining Landscape Connectivity of Low-cost Solutions for Multilayer Nets

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Abstract

Mode connectivity (Garipov et al., 2018; Draxler et al., 2018) is a surprising phenomenon in the loss landscape of deep nets. Optima—at least those discovered by gradient-based optimization—turn out to be connected by simple paths on which the loss function is almost constant. Often, these paths can be chosen to be piece-wise linear, with as few as two segments.

We give mathematical explanations for this phenomenon, assuming generic properties (such as dropout stability and noise stability) of well-trained deep nets, which have previously been identified as part of understanding the generalization properties of deep nets. Our explanation holds for realistic multilayer nets, and experiments are presented to verify the theory.

1 Introduction

Efforts to understand how and why deep learning works have led to a focus on the *optimization landscape* of the training loss. Since optimization to near-zero training loss occurs for many choices of random initialization, it is clear that the landscape contains many global optima (or near-optima). However, the loss can become quite high when interpolating between found optima, suggesting that these optima occur at the bottom of "valleys" surrounded on all sides by high walls. Therefore the phenomenon of *mode connectivity* (Garipov et al., 2018; Draxler et al., 2018) came as a surprise: optima (at least the ones discovered by gradient-based optimization) are connected by simple paths in the parameter space, on which the loss function is almost constant. In other words, the optima are not walled off in separate valleys as hitherto believed. More surprisingly, the paths connecting discovered optima can be piece-wise linear with as few as two segments.

Mode connectivity begs for theoretical explanation. One paper (Freeman and Bruna, 2016) attempted such an explanation for 2-layer nets, even before the discovery of the phenomenon in multilayer nets. However, they require the width of the net to be exponential in some relevant parameters. Others (Venturi et al., 2018; Liang et al., 2018; Nguyen et al., 2018; Nguyen, 2019) require special structure in their networks where the number of neurons needs to be greater than the number of training data points. Thus it remains an open

problem to explain mode connectivity even in the 2-layer case with realistic parameter settings, let alone for standard multilayer architectures.

At first sight, finding a mathematical explanation of the mode connectivity phenomenon for multilayer nets—e.g., for a 50-layer ResNet on ImageNet—appears very challenging. However, the glimmer of hope is that since the phenomenon exists for a variety of architectures and datasets, it must arise from some generic property of trained nets. The fact that the connecting paths between optima can have as few as two linear segments further bolsters this hope.

Strictly speaking, empirical findings such as in (Garipov et al., 2018; Draxler et al., 2018) do not show connectivity between *all* optima, but only for *typical* optima discovered by gradient-based optimization. It seems an open question whether connectivity holds for all optima in overparametrized nets. Section 5 answers this question, via a simple example of an overparametrized two-layer net, not all of whose optima are connected via low-cost paths.

Thus to explain mode connectivity one must seek generic properties that hold for optima obtained via gradient-based optimization on realistic data. A body of work that could be a potential source of such generic properties is the ongoing effort to understand the generalization puzzle of over-parametrized nets—specifically, to understand the "true model capacity". For example, Morcos et al. (2018) note that networks that generalize are insensitive to linear restrictions in the parameter space. Arora et al. (2018) define a *noise stability* property of deep nets, whereby adding Gaussian noise to the output of a layer is found to have minimal effect on the vector computed at subsequent layers. Such properties seem to arise in a variety of architectures purely from gradient-based optimization, without any explicit noise-injection during training—though of course using small-batch gradient estimates is an implicit source of noise-injection. (Sometimes training also explicitly injects noise, e.g. dropout or batch-normalization, but that is not needed for noise stability to emerge.)

Since resilience to perturbations arises in a variety of architectures, such resilience counts as a "generic" property for which it is natural to prove mode connectivity as a consequence. We carry this out in the current paper. Note that our goal here is not to explain every known detail of mode connectivity, but rather to give a plausible first-cut explanation.

First, in Section 3 we explain mode connectivity by assuming the network is trained via dropout. In fact, the desired property is weaker: so long as there *exists* even a *single* dropout pattern that keeps the training loss close to optimal on the two solutions, our proof constructs a piece-wise linear path between them. The number of linear segments grows linearly with the depth of the net.

Then, in Section 4 we make a stronger assumption of noise stability along the lines of Arora et al. (2018) and show that it implies mode connectivity using paths with 10 linear segments. While this assumption is strong, it appears to be close to what is satisfied in practice. (Of course, one could explicitly train deep nets to satisfy the needed noise stability assumption, and the theory applies directly to them.)

1.1 Related work

The landscape of the loss function for training neural networks has received a lot of attention. Dauphin et al. (2014); Choromanska et al. (2015) conjectured that local minima of multi-layer neural networks have similar loss function values, and proved the result in idealized settings. For linear networks, it is known (Kawaguchi, 2016) that all local minima are also globally optimal.

Several theoretical works have explored whether a neural network has spurious valleys (non-global minima that are surrounded by other points with higher loss). Freeman and Bruna (2016) showed that for a two-layer net, if it is sufficiently overparametrized then all the local minimizers are (approximately) connected. However, in order to guarantee a small loss along the path they need the number of neurons to be exponential in the number of input dimensions. Venturi et al. (2018) proved that if the number of neurons is larger than either the number of training samples or the intrinsic dimension (infinite for standard architectures), then the neural network cannot have spurious valleys. Liang et al. (2018) proved similar results for the binary classification setting. Nguyen et al. (2018); Nguyen (2019) relaxed the requirement on overparametrization, but still require the output layer to have more direct connections than the number of training samples.

Some other papers have studied the existence of spurious local minima. Yun et al. (2018) showed that in most cases neural networks have spurious local minima. Note that a local minimum need only have loss no

larger than the points in its neighborhood, so a local minimum is not necessarily a spurious valley. Safran and Shamir (2018) found spurious local minima for simple two-layer neural networks under a Gaussian input distribution. These spurious local minima are indeed spurious valleys as they have positive definite Hessian.

2 Preliminaries

Notations For a vector v, we use $\|v\|$ to denote its ℓ_2 norm. For a matrix A, we use $\|A\|$ to denote its operator norm, and $\|A\|_F$ to denote its Frobenius norm. We use $[n]$ to denote the set $\{1, 2, \ldots, n\}$. We use I_n to denote the identity matrix in $\mathbb{R}^{n\times n}$. We use $O(\cdot), \Omega(\cdot)$ to hide constants and use $\widetilde{O}(\cdot), \widetilde{\Omega}(\cdot)$ to hide poly-logarithmic factors.

Neural network In most of the paper, we consider fully connected neural networks with ReLU activations. Note however that our results can also be extended to convolutional neural networks (in particular, see Remark 1 and the experiments in Section 6).

Suppose the network has d layers. Let the vector before activation at layer i be x^i, $i \in [d]$, where x^d is just the output. For convenience, we also denote the input x as x^0. Let A_i be the weight matrix at i-th layer, so that we have $x^i = A_i\phi(x^{i-1})$ for $2 \le i \le d$ and $x^1 = A_1x^0$. For any layer i, $1 \le i \le d$, let the width of the layer be h_i. We use $[A_i]_j$ to denote the j-th column of A_i. Let the maximum width of the hidden layers be $h_{\max} := \max\{h_1, h_2, \ldots, h_{d-1}\}$ and the minimum width of the hidden layers be $h_{\min} := \min\{h_1, h_2, \ldots, h_{d-1}\}$.

We use Θ to denote the set of parameters of neural network, and in our specific model, $\Theta = \mathbb{R}^{h_1\times h_0} \times \mathbb{R}^{h_2\times h_1} \times \cdots \times \mathbb{R}^{h_d\times h_{d-1}}$ which consists of all the weight matrices $\{A_i\}$'s.

Throughout the paper, we use f_θ, $\theta \in \Theta$ to denote the function that is computed by the neural network. For a data set $(x, y) \sim \mathcal{D}$, the loss is defined as $L_{\mathcal{D}}(f_\theta) := \mathbb{E}_{(x,y)\sim\mathcal{D}}[l(y, f_\theta(x))]$ where l is a loss function. The loss function $l(y, \hat{y})$ is convex in the second parameter. We omit the distribution $\mathcal{D}$ when it is clear from the context.

Mode connectivity and spurious valleys Fixing a neural network architecture, a data set $\mathcal{D}$ and a loss function, we say two sets of parameters/solutions θ^A and θ^B are **ϵ-connected** if there is a path $\pi(t) : \mathbb{R} \to \Theta$ that is continuous with respect to t and satisfies: 1. $\pi(0) = \theta^A$; 2. $\pi(1) = \theta^B$ and 3. for any $t \in [0, 1]$, $L(f_{\pi(t)}) \le \max\{L(f_{\theta^A}), L(f_{\theta^B})\} + \epsilon$. If $\epsilon = 0$, we omit ϵ and just say they are connected.

If all local minimizers are connected, then we say that the loss function has the **mode connectivity property**. However, as we later show in Section 5, this property is very strong and is not true even for overparametrized two-layer nets. Therefore we restrict our attention to classes of low-cost solutions that can be found by the gradient-based algorithms (in particular in Section 3 we focus on solutions that are dropout stable, and in Section 4 we focus on solutions that are noise stable). We say the loss function has **ϵ-mode connectivity property** with respect to a class of low-cost solutions $\mathcal{C}$, if any two minimizers in $\mathcal{C}$ are ϵ-connected.

Mode connectivity is closely related to the notion of spurious valleys and connected sublevel sets (Venturi et al., 2018). If a loss function has all its sublevel sets ($\{\theta : L(f_\theta) \le \lambda\}$) connected, then it has the mode connectivity property. When the network only has the mode connectivity property with respect to a class of solutions $\mathcal{C}$, as long as the class $\mathcal{C}$ contains a global minimizer, we know there are no spurious valleys in $\mathcal{C}$.

However, we emphasize that neither mode connectivity or lack of spurious valleys implies any local search algorithm can efficiently find the global minimizer. These notions only suggest that it is unlikely for local search algorithms to get completely stuck.

3 Connectivity of dropout-stable optima

In this section we show that *dropout stable* solutions are connected. More concretely, we define a solution θ to be ϵ-dropout stable if we can remove a subset of half its neurons in each layer such that the loss remains